

Wave 2 Flavor Scaffolds Design

Wave 2 — Flavor Scaffolds Design

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Wave 2 — Flavor Scaffolds Design

Field	Value
Revision	1
Created	2026-05-21
Last modified	2026-05-21
Status	active
Status summary	Wave 2 of the master roadmap — 6 new flavor binaries (<code>sherald</code> , <code>cherald</code> , <code>bherald</code> , <code>rherald</code> , <code>iherald</code> , <code>scherald</code>) + <code>pherald</code> refactor. Shared CLI scaffold at <code>commons/cli/</code> . Three serving flavors (<code>cherald</code> + <code>sherald</code> + <code>iherald</code>). 28 §43 command stubs distributed across the 7 flavors. <code>e2e_bluff_hunt</code> invariants grow from 18 → ~33. Spec V3 r7 → r8 captures the design decisions.
Issues	none
Issues summary	—
Fixed	(n/a — design doc)
Continuation	

Field	Value
	After approval + spec update, invoke <code>superpowers:writing-plans</code> to author the Wave 2 implementation plan, then <code>superpowers:subagent-driven-development</code> per Universal §11.4.70 to dispatch task subagents.

Constitutional anchors

- **§107 end-user-usability covenant** — every Wave 2 flavor binary’s “done” is observable in `e2e_bluff_hunt.sh` (binary builds + version returns JSON + serve binds + healthz 200 + stub returns 501 with HRD pointer). Compile-only PASS is forbidden.
- **§11.4.70 subagent-driven default** — implementation proceeds via `superpowers:subagent-driven-development` (one subagent per task; spec + code review per task).
- **§11.4.74 catalogue-check** — before writing `commons/cli/`, search `vasic-digital` + `HelixDevelopment` for an existing Cobra-scaffold module. Disposition recorded in the implementation plan.
- **§11.4.73 spec versioning** — Wave 2 design changes ARE spec changes; V3 r7 → r8 captures them in the same commit cycle.
- **§3 + §11.4.79** — each new flavor is an own-org submodule conceptually (lives under Herald’s tree); included in CodeGraph index per the recently-landed mandate.

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Section 1 — Repo + module layout

```

Herald/
├── commons/                                (existing – extend with cli/
package)
│   └── cli/                                NEW – shared scaffold:
│       └── root.go                          NewRootCmd(branding, opts) –
Cobra root
│       └── version.go                       VersionCmd(branding) – JSON
build-info
│       └── serve.go                         ServeCmd(opts) – Gin server

```

```

with healthz/readyz/metrics
|   └─ routes.go           Route + HealthzHandler +
ReadyzHandler + MetricsHandler
|   └─ stubs.go           StubCmd(name, hrd,
description) - 501-style CLI
|   └─ cli_test.go        unit tests (table-driven)
└─ pherald/               (existing - REFACTORED to use
commons/cli)
└─ sherald/               NEW - System Herald (serving
daemon)
|   └─ go.mod · cmd/sherald/main.go (~30 LOC: brand + import
flavor stubs)
|   └─ internal/stubs/stubs.go 5 $43 stubs
└─ cherald/               NEW - Constitution Herald
(serving)
|   └─ go.mod · cmd/cherald/main.go
|   └─ internal/stubs/stubs.go 11 $43 stubs
|   └─ internal/http/routes.go /v1/compliance 501 stub (→
HRD-028)
└─ bherald/               NEW - Build Herald (CLI-only)
|   └─ go.mod · cmd/bherald/main.go
|   └─ internal/stubs/stubs.go 3 $43 stubs
└─ rherald/               NEW - Release Herald (CLI-
only)
└─ iherald/               NEW - Incident Herald
(serving)
|   └─ go.mod · cmd/iherald/main.go
|   └─ internal/http/routes.go /v1/webhooks/page 501 stub (→
HRD-024)
└─ scherald/               NEW - Scheduled-audit Herald
(CLI-only)

```

Module count: go.work grows 7 → 14 (+commons unchanged + 6 new flavors; pherald refactor stays in pherald/).

Layout rules: - cmd/<flavor>herald/main.go is ~30 LOC: declare branding, build root via cli.NewRootCmd(branding), register stubs, optionally call cli.ServeCmd(opts). - internal/stubs/stubs.go registers every \$43 command targeted at the flavor as a cli.StubCmd(...). - internal/http/routes.go (only for serving flavors) declares additional 501-stub routes per flavor.

Section 2 — Per-flavor identity, ports, serve surface, \$43 stub mapping

Flavor	Mission	Serves?	Default port	\$43 stubs (count)
pherald	Project Herald — multi-mirror push + submodule	✓	24791	6 (029 commit-push, 030 submodule-propagate,

Flavor	Mission	Serves?	Default port	§43 stubs (count)
s herald	propagation + project bindings			043 install-upstreams, 044 fetch-guard, 049 reopen, 053 pre-push)
	System Herald — destructive-op intercept + force-push gate + mem-budget watcher	✓ daemon	24793	5 (033 destructive-guard, 034 backup-snapshot, 040 constitution-pull, 046 force-push-gate, 056 mem-budget-watch)
	Constitution Herald — policy evaluator + creds scan + docs sync + composite gate	✓	24792	11 (036 creds-scan, 037 docs-sync, 038 script-docs-check, 039 fixed-align, 042 submanifest-verify, 048 fixed-summary-sync, 050 readme-sync, 051 composite-gate, 052 export, 054 spec-version-check, 055 catalogue-check)
b herald	Build Herald — CI/test bindings + test-tier verifier + evidence captureer	×	n/a	3 (035 evidence-capture, 041 test-tier-verify, 045 gate-retest — <i>045 owned by r herald; b herald gets gate-retest as cross-flavor alias</i>)

Flavor	Mission	Serves?	Default port	§43 stubs (count)
rherald	Release Herald — tag mirroring + changelog + installable-asset evidence	×	n/a	3 (031 tag-mirror, 032 changelog-generate, 045 gate-retest)
iherald	Incident Herald — credential-leak page-out + operator-blocked escalation	✓ paging webhooks	24794	0 (HRD-024 is §42 binding; its operator-visible surface is /v1/webhooks/page on the serve port)
sherald	Scheduled-audit Herald — periodic Status.md sweep + compliance digest	×	cron-driven n/a	1 (047 status-digest)

Stub total: 28 §43 stubs + 1 cross-flavor alias (045 shared between bherald + rherald) = 29 stub registrations. Matches §43 catalogue (HRD-029..HRD-056 = 28 unique HRDs).

Port allocation: 2479X block. Reserved: pherald 24791, cherald 24792, sherald 24793, iherald 24794. Free: 24795..24799 for future flavors.

Serve-surface routes per serving flavor

Common base (provided by commons/cli/routes.go): - GET /v1/healthz — 200 with {status:"ok",build:{version,go_version}} - GET /v1/readyz — 200 with {status:"ready"} - GET /metrics — Prometheus text + <flavor>_build_info gauge

Flavor-specific routes (501 stubs in Wave 2):

Flavor	Additional routes	Target HRD
pherald	POST /v1/events (existing 501)	HRD-016
cherald	GET /v1/compliance	HRD-028
sherald	GET /v1/safety_state (daemon status — open events, current mem%, last destructive-op log)	HRD-NNN follow-up (will track)
iherald		HRD-024

Flavor	Additional routes	Target HRD
	POST /v1/webhooks/ page (PagerDuty/ Opsgenie-compatible inbound)	

Branding (per spec §11.0 + §8.2)

Each flavor's `main.go` constructs a `commons.Branding{}` instance:

```
commons.Branding{
    Flavor:      "sherald",
    Prefix:      "s",                                // §8.2 per-flavor 3-
               letter prefix anchor
    DisplayName: "System Herald",
    DefaultPort: 24793,
    Mission:     "Host safety + destructive-op intercept + mem-
               budget watcher",
}
```

The `Branding` struct already exists in `commons/types.go` (HRD-009). `commons/cli.NewRootCmd(branding)` consumes `Branding` to populate Cobra's `Use/Short/Long/Version`.

Section 3 — `commons/cli/` package API surface

```
// Package cli is the shared scaffold for every Herald flavor
// binary.
// Per Universal §11.4.74 catalogue-check: extends nothing in the
// vasic-digital / HelixDevelopment orgs (no-match → vendor as
// Herald-internal package).
package cli

// NewRootCmd returns the top-level Cobra command for a flavor.
// The
// branding drives Use / Short / Long / Version. opts are
// functional
// options for callers that need to override defaults.
func NewRootCmd(branding commons.Branding, opts ...RootOpt)
    *cobra.Command

// VersionCmd is the `<flavor>herald version --json` subcommand.
// Returns
// canonical JSON with: binary, flavor, version, go_version, os,
// arch,
// commit_sha (from -ldflags), build_time. §107 anti-bluff: an
// empty
// version field is explicitly rejected by the e2e probe.
func VersionCmd(branding commons.Branding) *cobra.Command

// ServeCmd is the `<flavor>herald serve` subcommand for serving
// flavors.
```

```

// Builds a Gin engine with healthz + readyz + metrics + every
// flavor-
// specific route, binds on opts.Port (default =
// branding.DefaultPort),
// graceful-shutdown on SIGTERM.
func ServeCmd(opts ServeOpts) *cobra.Command

type ServeOpts struct {
    Branding commons.Branding
    Routes    []Route           // flavor-specific routes
                appended after base
    Logger    *logrus.Logger         // optional; defaults to a no-
                op logger
}

// Route declares one HTTP route. Wave 2 stubs return 501 with HRD
// pointer.
type Route struct {
    Method    string           // "GET", "POST", ...
    Path      string           // "/v1/compliance"
    Handler   gin.HandlerFunc
    Description string         // operator-readable summary
    HRD       string           // "HRD-028" for 501 stubs;
                "" for live routes
}

// StubCmd returns a Cobra subcommand that always returns 501-
// style error
// with the HRD pointer. Used by every flavor's internal/stubs/ to
// register
// §43 commands not yet implemented.
func StubCmd(name, hrd, description string) *cobra.Command

// HealthzHandler / ReadyzHandler / MetricsHandler are exposed so
// flavor-
// specific routes can compose them when needed.
func HealthzHandler(branding commons.Branding) gin.HandlerFunc
func ReadyzHandler(branding commons.Branding) gin.HandlerFunc
func MetricsHandler(branding commons.Branding) gin.HandlerFunc

```

Sub-package (only if needed): commons/cli/internal/build/ for build-info ldflags wiring (commit SHA + build time). Otherwise build info lives in commons/cli/version.go.

Dependencies: commons/cli/ imports commons/ (for Branding + types) + github.com/spf13/cobra + github.com/gin-gonic/gin + github.com/sirupsen/logrus. All already in pherald/go.mod; the refactor moves them into commons/go.mod.

Section 4 — e2e_bluff_hunt invariants (E19-E33)

Wave 2 adds **15 new invariants** to `scripts/e2e_bluff_hunt.sh`. Pre-Wave-2: 18 PASS. Post-Wave-2: **33 PASS** (with creds + container runtime), or 18 PASS + 15 SKIP cleanly.

Invariant	Probe	Target
E19	sherald version --json returns valid JSON with flavor=sherald + non-empty version/go_version	flavor build
E20	cherald version --json	flavor build
E21	bherald version --json	flavor build
E22	rherald version --json	flavor build
E23	iherald version --json	flavor build
E24	scherald version --json	flavor build
E25	sherald serve binds on :24793 + /v1/healthz returns 200 + status:ok + version	sherald serve
E26	cherald serve binds on :24792 + /v1/healthz returns 200	cherald serve
E27	iherald serve binds on :24794 + /v1/healthz returns 200	iherald serve
E28	cherald GET /v1/compliance returns 501 with HRD-028 pointer	cherald stub route
E29	sherald GET /v1/safety_state returns 501 with HRD-NNN pointer	sherald stub route
E30	iherald POST /v1/webhooks/page returns 501 with HRD-024 pointer	iherald stub route
E31	One representative §43 stub per flavor (e.g. sherald destructive-guard returns exit≠0 with “HRD-033” in stderr)	stub CLI exec

Invariant	Probe	Target
E32	All 3 serving flavors graceful-shutdown on SIGTERM (exit 0)	serve teardown
E33	pherald refactor regression — E1..E13 still PASS on the refactored binary	pherald continuity

Anti-bluff posture: each E19-E24 asserts non-empty `version` + `go_version` fields in the JSON. A `version: ""` PASS is explicitly rejected. E28-E30 assert the response body contains the literal `HRD-MNN` string (not just status 501). E31 asserts exit code $\neq 0$ AND `stderr` contains the HRD reference (catches a stub that exits 0 with “not implemented” silently).

Section 5 — §107 evidence contract per flavor

For each new flavor binary, the §107 PASS contract:

1. **Binary builds** — `go build ./<flavor>/...` produces an executable, captured via file `<binary>` MIME check.
2. **version --json returns canonical JSON** — parsed by python3 in the e2e probe, required fields asserted non-empty.
3. **<flavor> (no args) prints usage + exits 0** — proves Cobra root wires up correctly.
4. **For serving flavors:**
 - `serve` binds within 10s of invocation; subsequent `GET /v1/healthz` returns 200.
 - `GET /v1/readyz` returns 200 with `status:ready`.
 - `GET /metrics` returns `text/plain` with `<flavor>_build_info` gauge.
 - Flavor-specific 501 routes return body containing the literal `HRD-NNN`.
 - `SIGTERM` → graceful shutdown → exit 0 within 5s.
5. **Stub commands return 501-style error** — one representative stub per flavor exercised; exit code $\neq 0$; `stderr` contains `HRD-NNN` reference.

For the pherald refactor: all of E1-E13 (15 invariants from Plan 1 era) **MUST** stay green. The refactor introduces **NO** new e2e invariants — just changes the internals.

Section 6 — Test + build strategy

Unit tests (default `go test`): - `commons/cli/cli_test.go` — table-driven tests for `StubCmd` (exit code + `stderr` containment), `VersionCmd` (JSON shape), Route registration, branding plumbing. - Per-flavor `cmd/<flavor>herald/main_test.go` — minimal smoke (root command builds, has expected subcommands). - Per-flavor `internal/stubs/stubs_test.go` — verifies the expected §43 commands are registered with the right HRDs.

Integration tests (`//go:build integration`): - Per-serving-flavor: boot the binary as a subprocess, hit endpoints, verify response bodies. Same pattern as pherald's existing E7-E12. - Cross-flavor smoke: run `<flavor>herald version --json` for each of 6 flavors, verify exits + non-empty fields.

Build verification: - `go build ./...` across all 14 modules — must be clean before committing. - `gofmt -l` must be empty across all modified packages. - `go vet` must be clean.

Anti-bluff §1.1 paired mutations (new gates): - Strip the HRD pointer from one stub's error message → `e2e_bluff_hunt E31` MUST FAIL. - Make `version --json` return an empty version string → `E19` MUST FAIL. - Make sherald's `serve` bind on a random port instead of 24793 → `E25` MUST FAIL (with explicit port mismatch).

Each new gate gets its mutation test in `tests/test_wave2_mutation_meta.sh` per §1.1.

Spec impact (V3 r7 → r8)

Wave 2 changes to capture in spec V3 r8:

1. **§18 Flavors** — confirm 3-serving / 4-CLI-only split. Document default ports 24791..24794. Document the §43 stub distribution per flavor.
2. **§41 REST API surface** — add `cherald /v1/compliance` + `sherald /v1/safety_state` + `iherald /v1/webhooks/page` to the route catalogue (501 stubs in Wave 2; live in Waves 3/4).
3. **§44 Foundation implementation contract** — extend with a §44.M Wave 2 milestone section capturing the 6-flavor + pherald-refactor scope + the §107 evidence contract.
4. **§3.5 Branding + per-flavor identity** — formalize the per-flavor Branding instance pattern (Flavor, Prefix, DisplayName, DefaultPort, Mission).
5. **§11.4.74 Catalogue-Check** — document the `commons/cli/` no-match disposition in the §11.4.74 catalogue-checks subdirectory.

All five edits are mandatory per §11.4.73 spec-change rule.

Implementation phases

Per the master roadmap + this design:

1. **Phase 2a** — `commons/cli/` package + unit tests + Catalogue-Check doc (1 day).
2. **Phase 2b** — Refactor pherald to consume `commons/cli/` + verify E1-E13 still PASS (1 day).
3. **Phase 2c** — Scaffold 6 new flavors in parallel (1 day each, can run in 1-3 days with parallel subagents).
4. **Phase 2d** — Add E19-E33 invariants to `e2e_bluff_hunt.sh` + paired §1.1 mutation tests (half day).
5. **Phase 2e** — Update spec V3 r7 → r8 + regen siblings (half day).
6. **Phase 2f** — Atomic Issues→Fixed for the flavor-scaffold sub-HRDs (one per flavor — open HRD-092..097) + multi-mirror push (half day).

Total: 5-7 days with parallel subagents.

Open questions deferred to implementation

These are NOT resolved in this design; the implementation plan resolves them.

1. **Exact commons.Branding struct fields** — the existing struct from HRD-009 may not have `DefaultPort` or `Mission`. The implementation plan checks the current shape and either adds fields or carries Wave-2-specific fields in a `cli.FlavorMeta` wrapper.
2. **Whether commons/cli/ lives at commons/cli/ or commons/internal/cli/** — the latter prevents external Herald-consumer projects from importing it. Default: `commons/cli/` (exported) because the same scaffold pattern may be useful to flavors built by future Helix-stack consumers.
3. **sherald/v1/safety_state HRD assignment** — currently TBD. The plan opens HRD-098 (or appropriate number) for the live implementation of that route.
4. **Per-flavor smoke test pattern** — Cobra's `cmd.Execute()` from inside a `go test` requires capturing `stdout/stderr` via `cmd.SetOut()` / `cmd.SetErr()`. Plan documents the exact pattern.

Next step

After this design doc is committed + pushed: 1. Spec V3 r7 → r8 update lands in the same commit cycle per §11.4.73. 2. `superpowers:writing-plans` skill is invoked to author the Wave 2 implementation plan with bite-sized tasks. 3. `superpowers:subagent-driven-development` dispatches task subagents per Universal §11.4.70.